Notes: Agarwal and Ben-David (JFE, 2018)

Loan Prospecting and the Loss of Soft Information

Contents

1 Big picture

1.1 Question

Agarwal and Ben-David study what happens when a bank changes the job of small-business loan officers from passive screening to active loan prospecting. The question is:

> When loan officers are rewarded for generating loan volume, do they bring in new good borrowers, or do they expand lending by relaxing how they screen existing applications?

The paper is about incentives, information production, and delegated lending. The key object is not simply whether more loans are made. The deeper question is whether the bank's use of **hard information** and **soft information** changes when loan officers are paid to prospect.

1.2 Setting

The setting is the small-business lending unit of a large U.S. commercial bank in New England. The bank introduced a pilot compensation program in January 2005. About half of the loan officers were moved from fixed salary compensation to a compensation structure tied to loan prospecting and loan origination outcomes. The other half remained on the old compensation structure.

The treatment was tied to legacy human-resources systems inherited from past mergers. Loan officers associated with one system entered the new incentive plan; loan officers associated with the other system stayed on fixed salary. This creates a quasi-experimental treatment-control design.

1.3 The intervention

The pilot changed both the scope of the loan officer's job and the compensation scheme. Treated loan officers were expected to prospect for new business, not merely evaluate applications that arrived at the branch.

The compensation formula had three components:

- **Originated dollar amount:** 50% weight.
- **Number of loans:** 25% weight.
- **Decision turnaround time:** 25% weight.

Loan officers in the pilot received a lower fixed salary, equal to about 80% of their previous salary, plus a bonus based on this formula. The plan also included a quality screen: loan officers could not have unsatisfactory underwriting exceed 5% of total approvals if they wanted to be eligible for the incentive program.

1.4 Core empirical tension

There are two possible ways the program could raise loan volume.

Good prospecting channel. Loan officers could find new borrowers, improve the application pool, and expand lending without lowering credit quality.

Screening-distortion channel. Loan officers could increase approvals and loan sizes by giving more weight to observable hard information and less weight to negative soft information. In that case, loans may look similar on paper but perform worse ex post.

The paper's main result strongly favors the second channel.

1.5 Main result

The pilot increased lending volume, but not because treated loan officers attracted a larger or observably better application pool. Instead, the treatment expanded lending through approval decisions and larger loan amounts.

Key magnitudes:

- The number of approved and originated loans rose by about **31%**.
- Average approved and originated loan size rose by about **15%**.
- Applications handled by treated officers did **not** increase significantly in number or observable quality.
- Approved loans did **not** look worse using observable hard information.
- Actual default rates increased by about **1.2 to 1.3 percentage points**, roughly a **24%-28%** relative increase from the control-group default rate.

1.6 Economic mechanism

The mechanism is an informational shift. Treated loan officers relied more on hard information, such as external credit scores, and less on soft information, such as their private assessment of the borrower and business.

Interpretation. The important fact is not that the bank knowingly approved worse loans on observable dimensions. The important fact is that the approved loans looked similar on paper, but defaulted more. This means that the missing information was likely soft information that the loan officer previously used to reject risky borrowers.

1.7 Broader takeaway

The paper shows how a seemingly reasonable sales incentive can damage the informational role of bank lending. Loan officers are valuable partly because they collect and use soft information. If incentives make them act more like salespeople, the bank may lose exactly the information advantage that makes relationship lending valuable.

2 Data and measurement

2.1 Data source and sample

The data contain all small-business loan applications submitted to the New England division of the bank in 2004 and 2005. The authors observe both rejected and approved applications, which is crucial because the paper studies the approval process, not just originated loans.

Main sample objects:

- **Loan applications:** 30,268 applications.
- **Originated loans:** 11,164 originated loans.
- **Loan officers:** about 130 loan officers.
- **Loan officer-month observations:** 3,192 observations.
- **Time period:** 2004, the pre-treatment year, and 2005, the pilot year.

The data are unusually rich because they include applications, approval decisions, originated amounts, requested amounts, collateral, credit scores, internal risk ratings, interest rates, decision times, withdrawal behavior, loan officer characteristics, and realized default outcomes.

2.2 Treatment and control

The authors define the treatment using the bank's pilot rollout.

- **Group A:** loan officers who stayed on fixed salary in both 2004 and 2005.
- **Group B:** loan officers who were on fixed salary in 2004 and moved into the loan prospecting incentive plan in 2005.

The treatment indicator equals one only for applications handled by Group B loan officers in 2005:

$$\text{LoanProspecting}_{lt} = 1\{\text{loan officer } l \in \text{Group B}\} \times 1\{t = 2005\}.$$

Interpretation. The control observations are Group A in both years and Group B before treatment. The treated observations are Group B after the compensation change.

2.3 Loan approval process

The loan approval process begins when a small-business owner submits an application with a requested loan amount, proposed collateral, and supporting financial information. The loan officer verifies the application, gathers external credit information, interviews the applicant, and forms an internal risk assessment.

The decision uses both hard and soft information.

Hard information includes credit scores, collateral, requested loan size, leverage, and other variables recorded in the bank's systems.

Soft information includes private information collected by the loan officer through meetings, interviews, local knowledge, and judgment about the borrower's business prospects.

The local credit committee, usually the branch manager and loan officer, makes the decision. Unlike residential mortgage lending, the bank can approve a commercial loan for a different amount than the applicant requested.

2.4 Key outcome variables

- **Application approved:** indicator equal to one if the bank approves the application.
- **Loan originated:** indicator equal to one if the loan is actually originated.
- **Originated amount:** the dollar amount the bank lends.
- **Approved amount greater than requested:** indicator equal to one if the bank approves more than the borrower requested.
- **Turnaround time:** time between application submission and decision, measured in months.
- **Withdrawn:** indicator equal to one if an approved application is withdrawn by the borrower.
- **Default within 12 months:** indicator equal to one if the loan becomes at least 90 days past due within 12 months of origination.

2.5 Key credit-quality variables

- **Experian business score:** external business credit score. Higher means better credit quality.
- **Experian personal score:** external personal credit score of the applicant. Higher means better credit quality.
- **Internal risk rating:** loan officer's internal assessment, ranging from 1 to 10. Unlike Experian scores, higher internal rating means higher risk.
- **Loan-to-value ratio (LTV):** loan amount divided by collateral value.
- **Interest rate:** rate charged on the originated loan.

2.6 Descriptive facts

In the raw data, the treatment group in 2005 stands out along several dimensions.

- Loan origination probability is higher in the treatment group.
- Originated loan amounts are larger in the treatment group.
- The fraction of loans where the approved amount exceeds the requested amount rises sharply in the treatment group.
- Observable credit characteristics are not obviously worse for treated loans.

- Default rates are higher for treated loans.

Interpretation. These raw patterns already preview the central result: the treatment produced more and larger loans that looked acceptable on observable dimensions but performed worse afterward.

3 Models and empirical specifications

3.1 Baseline difference-in-differences model

The core empirical specification is a difference-in-differences design:

$$Y_{ilt} = \alpha + \beta \text{LoanProspecting}_{lt} + \Gamma' X_{ilt} + \mu_l + \tau_t + \delta_j + \varepsilon_{ilt}.$$

Here:

- Y_{ilt} is an outcome for application or loan i, handled by loan officer l, at time t.
- $\text{LoanProspecting}_{lt}$ equals one for treated loan officers in 2005.
- X_{ilt} is a vector of application or loan controls.
- μ_l are loan officer fixed effects.
- τ_t are month fixed effects.
- δ_j are industry fixed effects when the regression is at the application or loan level.

Interpretation. The coefficient β compares how outcomes changed for treated loan officers after the pilot relative to how outcomes changed for control loan officers over the same time period. Loan officer fixed effects mean identification comes from within-loan-officer changes over time.

3.2 Application-flow regressions

To test whether prospecting actually brought in more applications, the paper uses loan officer-month regressions such as:

$$\log(\text{Applications}_{lt}) = \alpha + \beta \text{LoanProspecting}_{lt} + \mu_l + \tau_t + \varepsilon_{lt}.$$

The same structure is used for average requested loan amount:

$$\log(\text{Avg requested amount}_{lt}) = \alpha + \beta \text{LoanProspecting}_{lt} + \mu_l + \tau_t + \varepsilon_{lt}.$$

Interpretation. If loan prospecting worked as intended, β should be positive: treated loan officers should bring more applications or larger requested loans. The paper finds no economically or statistically significant increase.

3.3 Application-quality regressions

The paper also asks whether treated loan officers attracted better applications. The specification is:

$$Q_{ilt} = \alpha + \beta \text{LoanProspecting}_{lt} + \Gamma' X_{ilt} + \tau_t + \delta_j + \varepsilon_{ilt},$$

where Q_{ilt} is an application characteristic such as requested amount, requested LTV, personal collateral, Experian business score, Experian personal score, or internal risk rating.

Interpretation. The coefficient β tests whether the treated application pool changed. The estimates are statistically insignificant, meaning the treatment did not bring in observably better or worse applications.

3.4 Ex-ante approval-likelihood model

To test whether treated loan officers attracted applications that were more likely to be approved, the authors first estimate an approval model using 2004 data:

$$\text{Approved}_i = f(\text{hard information}_i, \text{risk rating}_i, \text{loan characteristics}_i, \text{fixed effects}) + u_i.$$

They then use this model to predict approval likelihood for 2005 applications and compare predicted approval probabilities across treated and control groups.

Interpretation. This asks whether treated loan officers found applications that looked better according to the bank's pre-pilot approval rule. The answer is no.

3.5 Extensive-margin and intensive-margin regressions

The extensive-margin tests use loan officer-month data:

$$\log(\#\text{Loans}_{lt}) = \alpha + \beta \text{LoanProspecting}_{lt} + \mu_l + \tau_t + \varepsilon_{lt}.$$

The intensive-margin tests use average loan size:

$$\log(\text{Avg amount}_{lt}) = \alpha + \beta \text{LoanProspecting}_{lt} + \mu_l + \tau_t + \varepsilon_{lt}.$$

Interpretation. The extensive margin asks whether more loans were approved or originated. The intensive margin asks whether each loan was larger. The paper finds large positive effects on both margins.

3.6 Hard-information weighting in approval decisions

To examine the information-processing channel, the paper estimates approval regressions with interactions between treatment and credit-quality variables:

$$\text{Approved}_{ilt} = \alpha + \beta\text{LoanProspecting}_{lt} + \theta_1\text{HardInfo}_{ilt} + \theta_2(\text{LoanProspecting}_{lt} \times \text{HardInfo}_{ilt}) + \theta_3\text{SoftProxy}_{ilt} + \theta_4(\text{Loan}$$

Hard information is proxied by external Experian scores. Soft information is proxied by the internal risk rating, because that rating partly reflects the loan officer's private assessment.

Interpretation. A positive θ_2 means treated loan officers put more weight on hard information. A weakening of the internal-risk-rating effect means treated loan officers put less weight on the soft-information proxy.

3.7 Imputed expected-default model

The authors estimate expected default using 2004 loans:

$$\text{Default}_i = \alpha + \Gamma' X_i + \mu_l + \delta_j + \varepsilon_i,$$

where X_i includes observable loan characteristics such as requested amount, collateral, Experian scores, LTV, and LTV squared. The fitted value is the imputed predicted default likelihood:

$$\widehat{PD}_i = \widehat{\Pr}(\text{Default}_i = 1 \mid X_i, \mu_l, \delta_j).$$

The paper then tests whether $\widehat{PD}_i$ differs between treated and control loans in 2005.

Interpretation. This is a way to ask whether loans looked riskier ex ante using the bank's old observable-information model. The main finding is that they did not look riskier, even though they defaulted more later.

3.8 Actual-default regressions

The core default regression is:

$$\text{Default}_i = \alpha + \beta\text{LoanProspecting}_i + \Gamma' X_i + \mu_l + \tau_t + \delta_j + \varepsilon_i.$$

The dependent variable equals one if the loan becomes at least 90 days past due within 12 months of origination.

Interpretation. If the treatment only expands lending to equally good borrowers, β should be zero. The paper finds $\beta > 0$, meaning treated loans default more often.

3.9 Default-driver regressions

To understand why default rises, the paper interacts the treatment with variables measuring extensive- and intensive-margin expansion:

$$\text{Default}_i = \alpha + \beta\text{LoanProspecting}_i + \rho Z_i + \lambda(\text{LoanProspecting}_i \times Z_i) + \Gamma' X_i + \mu_l + \tau_t + \delta_j + \varepsilon_i.$$

The variable Z_i is one of the following:

- an approval residual, measuring unusually high approval relative to the old approval rule;
- an indicator for approved amount greater than requested amount;
- an abnormal LTV residual;
- an abnormal loan-size residual.

Interpretation. The interaction λ tells whether treated loans that are especially aggressive are more likely to default. The estimates show that both the extensive-margin expansion and aggressive loan terms help explain the default increase.

3.10 Credit-model breakdown regressions

Finally, the paper tests whether the old credit model still predicts default after treatment:

$$\text{Default}_i = \alpha + \beta \text{LoanProspecting}_i + \phi \widehat{PD}_i + heta(\text{LoanProspecting}_i \times \widehat{PD}_i) + \Gamma' X_i + \mu_l + \tau_t + \delta_j + \varepsilon_i.$$

Interpretation. In the control group, predicted default likelihood predicts actual default. In the treatment group, the predictive relation weakens or disappears, especially for loans with aggressive terms. The credit model breaks down because the loan population and loan terms move outside the range where the model was informative.

4 Identification and instruments

4.1 What identifies the estimates?

The paper does not use a traditional instrumental variable. Identification comes from the bank's controlled pilot and a difference-in-differences design.

The identifying comparison is:

> How did lending outcomes change from 2004 to 2005 for loan officers assigned to the new compensation system, relative to loan officers who stayed on fixed salary?

The treatment assignment is credible because it was based on legacy human-resources systems, not on loan officer performance, loan demand, borrower quality, or branch prospects.

4.2 Why the design is credible

Several features make the design strong.

1. **Quasi-random assignment.** Treatment was determined by the computer system used to administer loan officer compensation.

2. **No switching.** Loan officers were not allowed to switch between the two systems.
3. **Pre-treatment balance.** In 2004, Group A and Group B look similar in application characteristics, originated-loan characteristics, and loan officer salary.
4. **Loan officer fixed effects.** These absorb persistent differences across officers.
5. **Month fixed effects.** These absorb aggregate time shocks, seasonal effects, and macro conditions common to treated and control officers.
6. **Industry fixed effects.** These control for differences in borrower industries.
7. **No parallel securitization change.** The bank retained all small-business loans on balance sheet; the lending model did not shift toward selling or securitizing loans.

4.3 Why the result is not just loan demand

A natural alternative explanation is that treated branches experienced higher loan demand. The paper addresses this directly. Application volume did not rise significantly, requested loan amounts did not rise significantly, and observable application quality did not improve.

Interpretation. The loan expansion is not driven by more or better applications arriving at treated loan officers. It is driven by how treated loan officers handled the applications they received.

4.4 Why the result is not just lower formal standards

Another alternative explanation is that the bank explicitly lowered its hard-information lending standards. The evidence does not support this view. Approved treated loans do not have worse observable credit scores, internal risk ratings, or interest rates. Imputed predicted default based on observable characteristics does not increase for treated approved loans.

Interpretation. The change is subtler than a formal lowering of standards. The bank approved loans that looked acceptable by hard information but had worse unobserved risk.

4.5 Hawthorne and strategic-behavior concerns

Because loan officers knew they were being observed, the pilot could create Hawthorne effects. The authors examine time-series patterns in approval rates and loan sizes. The treatment effects appear immediately after the pilot and do not fade during 2005.

The paper also considers strategic behavior by loan officers. If treated officers wanted the pilot to succeed, they should have worked to bring in new good applications. If they wanted it to fail, they should have avoided increasing volume. Instead, the data show a different pattern: no new application flow, but more approvals, larger loans, and worse defaults. This looks like incentive-induced screening distortion rather than strategic sabotage or strategic polishing.

4.6 Remaining limitations

The setting is unusually clean, but it still has limits.

- The experiment comes from one bank, one region, and one small-business lending unit.
- Soft information is not directly observed; the conclusion is inferred from ex-post default patterns and changes in the weight placed on internal ratings.
- The treatment changes both job scope and compensation, so the estimates capture the combined effect of loan prospecting incentives.
- Default is measured within 12 months; longer-horizon loan performance is not the central outcome.

5 Tables and figures

5.1 Table 1: Summary statistics

Read: Table 1 splits the data by year and group: Group A in 2004, Group B in 2004, Group A in 2005, and Group B in 2005. Group B in 2005 is the treatment cell. The table separately reports loan applications, originated loans, and loan officer-month observations.

Important descriptive patterns:

- The application sample contains 30,268 applications.
- The originated-loan sample contains 11,164 loans.
- Group B in 2005 originates more loans: originated-loan probability is 46.6%, compared with 35.7% for Group A in 2005.
- Originated loan amount is larger in Group B in 2005: about $301,000, compared with about $253,000 for Group A in 2005.
- The share of loans where approved amount exceeds requested amount is 17.4% in the treatment cell, compared with roughly 3% in the control cells.
- Default within 12 months is 5.2% in the treatment cell, compared with around 4.2%-4.3% in the control cells.

Economic message: The treatment cell looks like a classic quantity-quality tradeoff: more loans and larger loans, but higher default. The key puzzle is that observable credit characteristics do not deteriorate enough to explain the default increase.

5.2 Table 2: Application volume and application quality

Read: Table 2 tests whether loan prospecting worked through the intended channel: attracting more or better applications.

Panel A shows that application volume and average requested amount do not rise significantly for treated loan officers. Panel B shows that observable application characteristics are not significantly different. Panel C shows that treated applications were not more likely to be approved according to a model estimated from pre-treatment data.

Table 1
Summary statistics.
The table presents summary statistics for the data used in the study. Panel A presents summary statistics for loan applications. Panel B presents summary statistics for the originated loans. Panel C presents summary statistics for data aggregated at the loan officer–month level. Variables are defined in Appendix A.

Panel A: Loan applications

	2004				2005			
	Group A (Control)		Group B (Control)		Group A (Control)		Group B (Treatment)	
# Applications:	6920		7996		7564		7788	
	Mean	St. dev.	Mean	St. dev.	Mean	St. dev.	Mean	St. dev.
Requested amount ($)	455,240	336,805	426,480	378,698	454,141	369,635	444,137	381,829
Personal collateral (0/1)	0.255	0.436	0.261	0.439	0.280	0.449	0.239	0.427
Requested LTV (%)	61.283	43.001	65.301	44.029	65.161	46.873	63.049	43.483
Experian business score (100–250)	200.863	72.228	195.884	75.868	195.988	75.273	200.359	68.471
Experian personal score (400–850)	731.847	70.305	725.405	68.063	725.908	74.394	728.057	76.723
Internal risk rating (1–10)	5.819	1.734	5.813	1.537	5.940	1.313	5.958	1.470
Time spent (months)	1.380	0.850	1.350	0.700	1.320	0.750	1.060	0.530
Application approved (0/1)	0.132	0.338	0.118	0.322	0.150	0.357	0.119	0.324
Withdrawn after being approved (0/1)	0.132	0.338	0.118	0.322	0.150	0.357	0.119	0.324

Panel B: Originated loans

	2004				2005			
	Group A (Control)		Group B (Control)		Group A (Control)		Group B (Treatment)	
# Originated loans:	2192		2548		2744		3680	
	Mean	St. dev.	Mean	St. dev.	Mean	St. dev.	Mean	St. dev.
Loan originated (0/1)	0.306	0.461	0.322	0.467	0.357	0.499	0.466	0.476
Requested amount ($)	302,074	305,891	302,966	301,933	303,082	306,939	302,224	317,073
Originated amount ($)	224,614	279,361	216,048	229,403	253,219	257,801	301,004	299,013
I(Amount approved > Amount requested)	0.035	0.025	0.032	0.028	0.032	0.027	0.174	0.073
Personal collateral (originated) (0/1)	0.270	0.409	0.280	0.403	0.300	0.420	0.250	0.404
Requested LTV (%)	79.060	20.930	78.440	19.280	79.030	17.040	78.520	18.400
Originated LTV (%)	72.986	31.477	76.237	30.899	74.901	33.105	77.033	26.049
Experian business score (100–250)	184.870	68.946	186.115	78.924	185.500	93.091	196.095	87.015
Experian personal score (400–850)	716.692	87.439	718.897	88.580	719.537	98.245	725.765	66.510
Time spent (months)	1.270	0.880	1.282	0.858	1.275	0.799	1.020	0.540
Internal risk rating (1–10)	5.230	1.840	5.380	1.520	5.440	1.300	4.930	1.530
Interest rate (%)	9.910	5.020	9.850	4.890	9.580	4.880	9.650	4.930
# Defaults	91		107		119		192	
Defaulted within 12 months (0/1)	0.042	0.199	0.042	0.201	0.043	0.204	0.052	0.222
log(Originated amount)–log(Requested amount)	−0.129	−0.039	−0.146	−0.117	−0.077	−0.075	0.014	0.104
Originated LTV-Requested LTV	−0.060	0.104	−0.022	0.116	−0.041	0.158	0.007	0.080
Residual from leverage regression	0.003	0.034	0.003	0.033	0.004	0.032	0.007	0.032
Residual from loan size regression	0.004	0.038	0.003	0.040	0.004	0.042	0.071	0.039

Panel C: Loan officer–month data

	2004				2005			
	Group A (Control)		Group B (Control)		Group A (Control)		Group B (Treatment)	
# Loan officers:	68		65		65		65	
	Mean	St. dev.	Mean	St. dev.	Mean	St. dev.	Mean	St. dev.
log(Application avg amount) ($k)	5.582	5.336	5.382	5.352	5.587	5.349	5.399	5.534
log(Approved avg amount) ($k)	5.293	5.562	5.296	5.485	5.290	5.433	5.525	5.661
log(Originated avg amount) ($k)	5.281	5.394	5.299	5.307	5.294	5.374	5.551	5.446
log(# Applications)	3.794	1.885	3.795	1.884	3.799	1.865	3.812	1.842
log(# Approved loans)	3.378	1.858	3.399	1.878	3.381	1.840	3.705	1.819
log(# Originated loans)	3.373	1.861	3.396	1.861	3.391	1.834	3.816	1.840
Salary ($)	43,292	32,941	43,023	32,114	43,139	32,327	47,305	32,672
log(Salary) ($)	4.567	4.555	4.583	4.544	4.608	4.601	4.660	4.521
Tenure (years)	3.1	2.6	3.1	2.6	3.0	2.6	3.1	2.6
Male (%)	62.9		68.0		64.5		66.9	

N(loan officer–month) = 3192.

Table 2
Volume and quality of applications.
The table presents an analysis of the loan application volume and characteristics. Panel A uses a sample at the loan officer–month level and explores whether the dollar volume and the number of applications are different for applications made to loan officers who engaged in loan prospecting. Panel B tests whether the characteristics of loan applications are different for applications made to loan officers who engage in loan prospecting. Panel C tests whether loan applications received by the loan prospecting group were more likely to be approved based on characteristics. Panels B and C use a sample of applications in the years 2004–2005, and 2005, respectively. All regressions are ordinary least squares regressions. Variables are defined in Appendix A. In Panel A, standard errors are clustered at the month level. In Panels B and C, standard errors are clustered at the loan officer level. Standard errors are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% level, respectively.

Panel A: Loan application volume in treated and control groups

Dependent variable:	All applications (loan officer–month)			
Denoted in:	log(Avg requested amount) ($)		log(# Applications)	
	(1)	(2)	(3)	(4)
Loan prospecting (0/1)	0.019	0.013	0.001	0.007
	(0.029)	(0.030)	(0.013)	(0.025)
Loan officer fixed effects	No	No	No	No
Month fixed effects	No	Yes	No	Yes
Observations	3192	3192	3192	3192
Adj. R^2	0.07	0.13	0.06	0.08

Panel B: Characteristics of loan applications

Dependent variable:	log(Requested amount)	Requested LTV	Personal collateral	Experian business score	Experian personal score	Internal risk rating
	(1)	(2)	(3)	(4)	(5)	(6)
Loan prospecting (0/1)	0.016	0.026	0.014	7.146	3.976	0.043
	(0.064)	(0.183)	(0.056)	(5.871)	(5.068)	(0.138)
Loan officer fixed effects	No	No	No	No	No	No
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Month fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	30,268	30,268	30,268	30,268	30,268	30,268
Adj. R^2	0.07	0.06	0.09	0.06	0.05	0.07

Panel C: Are loan applications in the treated group more likely to be approved?

Sample: Dependent variable:	2005 Applications			
	Application ex-ante probability of approval			
	(1)	(2)	(3)	(4)
Loan prospecting (0/1)	0.008	−0.007	0.004	0.001
	(0.018)	(0.019)	(0.019)	(0.019)
Experian business score			0.071***	0.076***
			(0.019)	(0.020)
Experian personal score			0.063***	0.094***
			(0.016)	(0.016)
Internal risk rating			−0.103**	−0.084**
			(0.026)	(0.025)
log(Requested amount)			−0.049	−0.030
			(0.008)	(0.008)
Personal collateral (0/1)			0.070	0.050
			(0.086)	(0.087)
Requested LTV			−0.026***	−0.028***
			(0.003)	(0.004)
Requested LTV^2			−0.120***	−0.117***
			(0.019)	(0.018)
Loan officer fixed effects	No	Yes	No	Yes
Industry fixed effects	Yes	Yes	Yes	Yes
Month fixed effects	Yes	Yes	Yes	Yes
Observations	15,352	15,352	15,352	15,352
Adj. R^2	0.14	0.17	0.15	0.17

Economic message: The treatment did not expand the bank's reach to new, better borrowers. Loan officers did not generate a meaningfully different application pool. This is the first major clue that the later increase in lending volume must come from the approval process, not from borrower inflow.

5.3 Figure 1: Approval residuals over time

Read: Figure 1 plots residual approval rates for Group A and Group B across months. The treated group shows a sharp increase immediately after the pilot begins in January 2005.

Economic message: The timing is important. Approval behavior changes right when the incentive plan starts, supporting the interpretation that the compensation change altered loan officer behavior.

5.4 Figure 2: Average originated loan amount over time

Read: Figure 2 plots average originated loan amounts by group and month. Group B's average originated loan size increases sharply in 2005 and stays elevated.

Economic message: The treatment affects the intensive margin as well as the extensive margin. Treated officers do not merely approve more loans; they also approve larger loans.

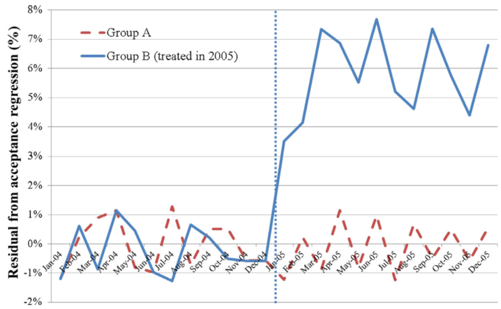

Fig. 1. Approval rate (residual) over time and across groups.
The chart shows the average residual from the approval regression (see Appendix C). The sample includes all applications in the years 2004–2005. T residuals are averaged within group (Groups A and B) and month. Note that whereas the regression uses only the control sample, the residuals calculated for the entire sample and therefore do not have a mean of zero.

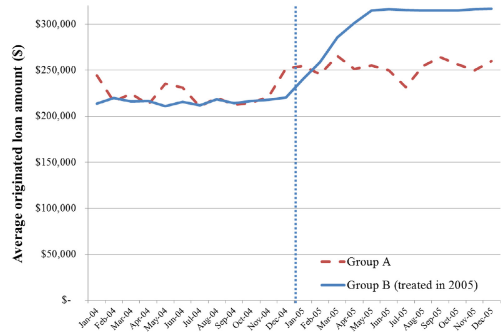

Fig. 2. Average originated loan amount over time and across groups.
The chart shows the average loan size. Loan sizes are averaged within group (Groups A and B) and month. The sample includes all originated loans in th years 2004–2005.

5.5 Table 3: Direct effects on approval, origination, loan size, speed, and withdrawals

Read: Table 3 estimates the direct treatment effects. Panel A shows that loan prospecting increases the number of approved and originated loans by about 30%-31%. It also increases average approved and originated amounts by about 14%-15%. Panel B shows that treated loan officers make decisions faster and that approved applications are less likely to be withdrawn.

Economic message: The pilot succeeds on the bank's short-run operating metrics: more loans, larger loans, faster decisions, and fewer withdrawals. But these are exactly the metrics in the compensation formula, so the results also show strong incentive responsiveness.

5.6 Table 4: Hard information and observed credit quality

Read: Table 4 asks whether treated officers approved loans with worse observable characteristics. Panel A shows that approved loans in treatment and control groups have statistically similar Experian scores, internal risk ratings, and interest rates. Panel B shows that treated approvals place more weight on external credit scores, while the role of internal risk rating weakens. Panel C shows that imputed predicted default does not increase for approved treated applications.

Economic message: Table 4 is central. The bank did not obviously approve worse loans according to observable variables. Instead, the treatment changed the information weights in the approval decision. Hard information became more important, and soft information became less important.

5.7 Table 5: Default rates

Read: Table 5 regresses default within 12 months on the treatment indicator and controls. Treated loans default more. The coefficient is about 1.2 to 1.3 percentage points depending on specification. Relative to a base default rate of about 4.3%, this is a large increase.

Economic message: The increase in default is the evidence that negative soft information was missed or ignored. If the treated loans were truly similar in all relevant dimensions, default should not increase.

Table 3
Direct effects of loan prospecting.
The table presents an analysis of the effects of loan prospecting: loan approval rates, loan origination rates, and average loan size (Panel A); turnaround time and withdrawal rate (Panel B); and default rate (Panel C). The sample used in Panel A is of loan officer-month observations. The sample used in Panel B, columns (1)–(4) is of all applications, and in columns (5)–(8) is of all approved loans. All regressions are ordinary least squares regressions. Variables are defined in Appendix A. Standard errors, reported in parentheses, are clustered at the month level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% level, respectively.

Panel A: Effects on the extensive and intensive margins of loan approval and origination

Dependent variable:	log(#Loans)				log(Avg amount) ($)			
Sample:	Approved		Originated		Approved		Originated	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Loan prospecting (0/1)	0.282***	0.313***	0.276***	0.305***	0.147***	0.149***	0.144***	0.145***
	(0.058)	(0.051)	(0.046)	(0.038)	(0.048)	(0.040)	(0.046)	(0.043)
Loan officer fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Month fixed effects	No	Yes	No	Yes	No	Yes	No	Yes
Observations	3192	3192	3192	3192	3192	3192	3192	3192
Adj. R^2	0.15	0.17	0.19	0.20	0.13	0.15	0.16	0.17

Panel B: Turnaround time and withdrawal rate

Dependent variable:	#Months between application and decision				I(Application withdrawn)	
Sample:	Applications		Approved loans		Approved loans	
	(1)	(2)	(3)	(4)	(5)	(6)
Loan prospecting (0/1)	−0.104*	−0.100**	−0.157**	0.166***	−0.057**	−0.068**
	(0.056)	(0.050)	(0.062)	(0.064)	(0.028)	(0.031)
Experian business score	−0.048***	−0.042***	−0.051***	−0.049***	0.048***	0.039**
	(0.017)	(0.016)	(0.019)	(0.017)	(0.016)	(0.016)
Experian personal score	−0.042***	−0.037**	−0.046***	−0.043***	0.039***	0.028**
	(0.014)	(0.014)	(0.016)	(0.014)	(0.011)	(0.014)
log(Requested amount)	0.019***	0.017***	0.021***	0.020***	−0.017***	−0.020***
	(0.007)	(0.006)	(0.007)	(0.007)	(0.005)	(0.004)
Personal collateral (0/1)	−0.032	−0.030	−0.038	−0.034	0.031	0.025
	(0.063)	(0.060)	(0.066)	(0.065)	(0.062)	(0.057)
Requested LTV	0.016***	0.016***	0.019***	0.017***	−0.014***	−0.014***
	(0.003)	(0.003)	(0.003)	(0.003)	(0.002)	(0.002)
Requested LTV^2	0.068***	0.068***	0.079***	0.070***	−0.062***	−0.042***
	(0.016)	(0.014)	(0.016)	(0.016)	(0.013)	(0.014)
Loan officer fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Month fixed effects	No	Yes	No	Yes	No	Yes
Observations	30,268	30,268	14,359	14,359	14,359	14,359
Adj. R^2	0.18	0.19	0.21	0.22	0.07	0.09

to a rate of about 13% in the control (Panel A of
. This effect is potentially due to a combination of

5.1. Hard information used in approving applications

Table 4
Did the Bank approve applications of lower observed credit quality?.
The table presents regressions testing whether the lending standards of the Bank changed along with the change in the scope of loan officer activities. Panel A tests whether a significant difference exists in the credit quality of loans that were approved in the treatment and control groups. Panel A uses only 2005 data, because the calculation of expected default uses 2004 data. Panel B presents an analysis of the effects of loan prospecting on the approval of loans with respect to credit quality. Panel C tests whether a difference exists in the imputed expected default likelihood of loans that were approved by the treatment and control groups. Imputed predicted default probability is the predicted value from a regression of the default indicator on observable characteristics using the 2004 applications data. Panels B and C use a sample of all applications in 2004–2005. All regressions are ordinary least square regressions. Variables are defined in Appendix A. Standard errors, reported in parentheses, are clustered at the loan officer level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% level, respectively.

Panel A: Characteristics of approved applications in the treatment and control groups

Sample:	All approved applications			
Dependent variable:	Experian business score	Experian personal score	Internal risk rating	Interest rate
	(1)	(2)	(3)	(4)
Loan prospecting (0/1)	5.333	3.129	0.045	0.007
	(5.880)	(5.286)	(0.139)	(0.047)
log(Approved amount)	0.041**	0.045**	0.041***	0.049***
	(0.016)	(0.020)	(0.016)	(0.019)
Personal collateral (0/1)	0.018***	0.020***	0.014***	0.022***
	(0.006)	(0.006)	(0.005)	(0.007)
Approved LTV	0.037***	0.040***	0.029**	0.041**
	(0.012)	(0.015)	(0.012)	(0.016)
Approved LTV^2	0.007***	0.008***	0.006***	0.007***
	(0.002)	(0.002)	(0.002)	(0.002)
Loan officer fixed effects	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes
Month fixed effects	Yes	Yes	Yes	Yes
Observations	14,359	14,359	14,359	14,359
Adj. R^2	0.36	0.28	0.27	0.18

Panel B: Focusing on hard information

Sample:	All applications					
Dependent variable:	Application approved (0/1)					
	(1)	(2)	(3)	(4)	(5)	(6)
Loan prospecting (0/1)	0.074***	0.084***	0.049***	0.038**	0.037**	0.040**
	(0.019)	(0.019)	(0.018)	(0.016)	(0.017)	(0.019)
× Experian business score			0.019***	0.015***	0.014***	0.013**
			(0.003)	(0.001)	(0.005)	(0.005)
× Experian personal score			0.005	0.008**	0.004***	0.008
			(0.003)	(0.003)	(0.001)	(0.005)
× Internal risk rating			0.016	0.015	0.016	0.016
			(0.014)	(0.011)	(0.012)	(0.011)
Experian business score					0.003	0.016***
					(0.010)	(0.006)
Experian personal score					0.006**	0.013***
					(0.003)	(0.003)
Internal risk rating					−0.026***	−0.024***
					(0.006)	(0.005)
log(Requested amount)					0.005***	0.004
					(0.001)	(0.004)
Personal collateral (0/1)					0.011	0.008
					(0.037)	(0.015)
Requested LTV					−0.008***	−0.002
					(0.000)	(0.002)
Requested LTV^2					−0.032***	−0.018***
					(0.005)	(0.003)
Loan officer fixed effects	No	Yes	No	Yes	No	Yes
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Month fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	30,268	30,268	30,268	30,268	30,268	30,268
Adj. R^2	0.14	0.14	0.15	0.15	0.15	0.15

(continued on next page)

Table 5
Default rates.
The table presents an analysis of the default rates following the loan prospecting experiment. The sample includes all originated loans in 2004–2005. All regressions are ordinary least squares regressions. Variables are defined in Appendix A. Standard errors, reported in parentheses, are clustered at the month level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% level, respectively.

Sample:	All originated loans			
Dependent variable:	Defaulted within 12 months (0/1)			
	(1)	(2)	(3)	(4)
Loan prospecting (0/1)	0.013***	0.006***	0.005*	0.007***
	(0.004)	(0.002)	(0.003)	(0.002)
× I(Amount approved > Amount requested)		0.018***		
		(0.005)		
× log(Originated loan amount) (residual)			0.013**	
			(0.005)	
× Originated LTV (residual)				0.013***
				(0.005)
Controls	Yes	Yes	Yes	Yes
Loan officer fixed effects	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes
Month fixed effects	Yes	Yes	Yes	Yes
Observations	11,164	11,164	11,164	11,164
Adj. R^2	0.21	0.21	0.22	0.23

Table 6
Default drivers: extensive and intensive margins.
The table explores the relation between loan amount and loan performance. The sample includes all originated loans in 2004–2005. All regressions are ordinary least square regressions. Variables are defined in Appendix A. Standard errors, reported in parentheses, are clustered at the loan officer level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% level, respectively.

Sample:	All originated loans				
Dependent variable:	Defaulted within 12 months (0/1)				
	(1)	(2)	(3)	(4)	(5)
Loan prospecting (0/1)	0.013***	0.067***	0.055***	0.066***	0.030***
	(0.004)	(0.003)	(0.002)	(0.002)	(0.002)
× Loan approved (residual)		0.040***			0.036***
		(0.015)			(0.014)
× I(Amount approved > Amount requested)			0.038***		0.034***
			(0.010)		(0.011)
× Originated LTV (residual)				0.034**	0.034**
				(0.013)	(0.015)
Loan approved (residual)		0.027**			0.026*
		(0.014)			(0.015)
I(Amount approved > Amount requested)			0.028**		0.026**
			(0.012)		(0.012)
Originated LTV (residual)				0.037**	0.035**
				(0.015)	(0.014)
log(Originated loan amount)	0.077***	0.082***	0.029	0.078***	0.007
	(0.022)	(0.027)	(0.022)	(0.025)	(0.019)
Personal collateral (0/1)	−0.038	−0.059	−0.055	−0.045	−0.043
	(0.033)	(0.043)	(0.034)	(0.035)	(0.028)
Experian business score	−0.001	−0.001*	−0.001	−0.001	−0.001
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Experian personal score	−0.001	−0.001	−0.001	−0.001	−0.001
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Originated LTV	0.014**	0.013**	0.016**		0.012***
	(0.006)	(0.006)	(0.007)		(0.004)
Originated LTV^2	0.035***	0.036***	0.033***		0.031***
	(0.002)	(0.002)	(0.002)		(0.002)
Interest rate (%)	0.038***	0.051***	0.048***	0.039***	0.044***
	(0.013)	(0.012)	(0.014)	(0.012)	(0.019)
Loan officer fixed effects	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes	Yes
Month fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	11,164	11,164	11,164	11,164	11,164
Adj. R^2	0.21	0.23	0.21	0.23	0.26

rocess and the composition of the loan portfo-
ighout the process, the Bank did not change its
approval criteria and credit model, but it did al-
ficantly larger loan amounts. Thus, an important
is how the Bank's predictive credit model fared.
words, could the Bank have predicted the in-
default? A parallel could be drawn with the res-
nortgage market during the early 2000s. There,
ctive credit models were applied to a new pop-
ubprime borrowers) with more aggressive lend-
s. Rajan et al. (2015) show the imputed predic-
t models that had been fitted to borrowers with
dit lost their predictive power once they were ap-
he subprime population.
ank's old credit model may have become obso-
to a combination of two factors. First, as shown
n 5.1, the Bank's approval process began to rely
avily on hard information. The combination of
s of information may have previously provided a
een for borrowers, whereas relying primarily on
rmation may have missed crucial details about
s. Second, loan terms under the treatment group
became significantly more aggressive. The sensitivity
performance to factors such as loan amounts and levera
might not be linear. Because the Bank was using a cre
model that was fit on data with a certain range of amour
and leverages, the model may not have been applicable
loans with more aggressive terms.

To test the proposition that the Bank's predictive moc
became defunct and to explore the reasons why, we co
duct three tests. We have seen that the Bank began to re
more heavily on hard information in the approval de
sion. The first test (Panel A of Table 7) therefore examin
how well observable information predicts ex-post defa
in the treatment versus the control group. The table sho
the different types of information (Experian business a
personal information as well as the internal risk ratin
that predict default for the entire sample. However, for t
treatment, these variables lose their predictive ability.

The second test offers a more elaborated specificati
of a default predictor. As in Table 4, we use the predict
default (based on the 2004 data) to explain ex-post d
fault of loans originated in 2005. For each originated lo
in 2005, we compute the predicted default likelihood. V

5.8 Table 6: Default drivers

Read: Table 6 studies which treated loans drive the higher default rate. It uses measures of unusual approval and aggressive loan terms: approval residuals, approval above requested amount, abnormal loan size, and abnormal LTV. The interaction terms are positive and significant.

Economic message: The default increase is not random. It is concentrated exactly where the incentives should bite: loans that were approved despite being unusual under the old rule, loans where the approved amount exceeds the requested amount, and loans with aggressive leverage.

5.9 Table 7: Breakdown of the credit model

Read: Table 7 tests whether the bank's old credit model still predicts default. In the control group, predicted default likelihood is informative about actual default. In the treatment group, predicted default loses power. The breakdown is strongest for loans with aggressive terms.

Economic message: Credit models are not structural laws. They are estimated under a given screening process. When incentives change the loan approval process, the distribution of originated loans changes, and an old model can stop working.

Table 7

Performance driver: breakdown of the credit model.

The table tests the hypothesis that the Bank's credit model broke down during the loan prospecting experiment. The sample includes all originated loans in 2004–2005. Panel A tests whether default is correlated with different information types. Panel B tests whether default is correlated with predicted default in the control and in the treatment groups. All regressions are ordinary least square regressions. Variables are defined in Appendix A. Standard errors, reported in parentheses, are clustered at the loan officer level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% level, respectively.

Panel A: Information types and default

Sample:	All originated loans					
Dependent variable:	Defaulted within 12 months (0/1)					
	(1)	(2)	(3)	(4)	(5)	(6)
Loan prospecting (0/1)	0.014***	0.012***	0.012**	0.012**	0.049***	0.006**
	(0.004)	(0.003)	(0.002)	(0.002)	(0.002)	(0.002)
× Experian business score			0.030***	0.021***	0.029***	0.017*
			(0.004)	(0.006)	(0.006)	(0.010)
× Experian personal score			0.011	0.014***	0.009*	0.011
			(0.008)	(0.005)	(0.005)	(0.009)
× Internal risk rating			0.026***	0.019**	0.031***	0.032***
			(0.006)	(0.009)	(0.009)	(0.009)
Experian business score			−0.020	−0.018	−0.020	−0.019
			(0.015)	(0.014)	(0.016)	(0.014)
Experian personal score			−0.016**	−0.017*	−0.017**	−0.018*
			(0.008)	(0.010)	(0.008)	(0.010)
Internal risk rating			0.037***	0.032	0.039***	0.033
			(0.014)	(0.020)	(0.014)	(0.021)
log(Requested amount)					0.014***	0.016***
					(0.003)	(0.005)
Personal collateral (0/1)					−0.021	−0.024
					(0.048)	(0.042)
Requested LTV					0.010***	0.010***
					(0.002)	(0.002)
Requested LTV^2					0.050***	0.047***
					(0.009)	(0.008)
Loan officer fixed effects	No	Yes	No	Yes	No	Yes
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Month fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	11,164	11,164	11,164	11,164	11,164	11,164
Adj. R^2	0.21	0.21	0.23	0.25	0.26	0.27

Panel B: Predicted default likelihood and actual default

Sample:	2005 Originated loans (Group A)	2005 Originated loans (Group B)	All 2005 Originated loans
Dependent variable:	Default (0/1)	Default (0/1)	Default (0/1)
	(1)	(2)	(3)
Loan prospecting (0/1)			0.010***
			(0.004)
× Predicted default likelihood			−0.015***
			(0.006)
× Internal risk rating			0.010**
			(0.005)
× Interest rate			−0.019***
			(0.003)
Predicted default likelihood	0.022**	0.003	0.022**
	(0.009)	(0.006)	(0.009)
Internal risk rating	−0.015*	−0.002	−0.014*
	(0.008)	(0.005)	(0.008)
Interest rate	0.043***	0.024***	0.046***
	(0.004)	(0.004)	(0.003)
Loan officer fixed effects	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes
Month fixed effects	Yes	Yes	Yes
Observations	2744	3680	6424
Adj. R^2	0.09	0.09	0.10

(continued on next page)

Table 8

Loan officer characteristics and loan origination.

The table tests the relation between loan officer characteristics (tenure, gender) and the loan origination process. The sample used in Panel A, columns (1)–(4), includes all applications in 2004–2005. The sample used in Panel A, columns (5)–(8), includes all approved loans in 2004–2005. Panel A tests the relation between the likelihood of loan approval and loan officer characteristics, shown in columns (1)–(4), and the relation between loan amounts and loan officer characteristics, shown in columns (5)–(8). Panel B tests the relation between loan officers' compensation and loan officer characteristics. The sample in Panel B consists of loan officer-month observations. All regressions are ordinary least square regressions. Variables are defined in Appendix A. Standard errors, reported in parentheses, are clustered at the loan officer level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% level, respectively.

Panel A: Loan officer characteristics and decision-making

Dependent variable:	Approved loan (0/1)				log(Loan amount)			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Loan prospecting (0/1)	0.054***	0.033***	0.041***	0.052***	0.103***	0.058***	0.032***	0.023**
	(0.018)	(0.006)	(0.002)	(0.006)	(0.010)	(0.010)	(0.011)	(0.010)
× Tenure above median (0/1)		0.025**		0.005*		0.051**		0.054***
		(0.009)		(0.003)		(0.025)		(0.017)
× Male (0/1)			0.020***	0.003			0.070***	0.050**
			(0.001)	(0.003)			(0.023)	(0.018)
Tenure above median (0/1)		0.003		0.004		0.002		0.003
		(0.002)		(0.013)		(0.012)		(0.002)
Male (0/1)			0.002	0.005**			0.005**	0.001
			(0.002)	(0.002)			(0.002)	(0.004)
Experian business score	0.003	0.003	0.004	0.006***	0.001	0.002	0.004	0.007***
	(0.002)	(0.005)	(0.004)	(0.002)	(0.002)	(0.003)	(0.009)	(0.000)
Experian personal score	0.005***	0.008***	0.010***	0.008***	0.003***	0.001	0.004**	0.006***
	(0.000)	(0.001)	(0.002)	(0.000)	(0.001)	(0.001)	(0.002)	(0.000)
Internal risk rating	0.008***	0.004	0.005	0.003	0.006	0.004***	0.002	0.001
	(0.000)	(0.008)	(0.008)	(0.007)	(0.007)	(0.000)	(0.009)	(0.002)
log(Requested amount)	0.000	0.001	0.001	0.001	0.877***	0.857***	0.878***	0.860***
	(0.000)	(0.004)	(0.001)	(0.004)	(0.123)	(0.122)	(0.121)	(0.120)
Personal collateral (0/1)	0.002	0.005	0.001	0.005	0.010	0.005	0.007	0.002
	(0.026)	(0.004)	(0.025)	(0.014)	(0.018)	(0.008)	(0.028)	(0.001)
Requested LTV	0.0000	−0.002	−0.001***	−0.002***	−0.006***	0.000*	−0.005***	−0.002***
	(0.000)	(0.001)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Requested LTV^2	−0.001	−0.006***	−0.003	−0.008***	−0.002***	−0.003	−0.004**	−0.001
	(0.002)	(0.002)	(0.004)	(0.002)	(0.000)	(0.003)	(0.002)	(0.001)
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Month fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	30,268	30,268	30,268	30,268	14,916	14,916	14,916	14,916
Adj. R^2	0.28	0.28	0.29	0.29	0.41	0.41	0.42	0.42

Panel B: Loan officer characteristics and compensation

Dependent variable:	log(Compensation) ($)			
	(1)	(2)	(3)	(4)
Loan prospecting (0/1)	0.082***	0.075***	0.066***	0.060***
	(0.020)	(0.014)	(0.016)	(0.015)
× Tenure above median (0/1)		0.025**		0.021**
		(0.011)		(0.011)
× Male (0/1)			0.036***	0.038***
			(0.011)	(0.010)
Tenure above median (0/1)		0.006		0.005
		(0.014)		(0.014)
Male (0/1)			0.007	0.006
			(0.012)	(0.011)
Month fixed effects	Yes	Yes	Yes	Yes
Observations	3192	3192	3192	3192
Adj. R^2	0.54	0.57	0.69	0.70

Panel A: Loan applications in groups A and B in 2004

Dependent variable:	log(Requested amount)	Requested LTV (%)	Personal collateral	Experian business score	Experian personal score	Internal risk rating	Time spent	Application withdrawn	log(Loan officer salary ($k))
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Group B (to be treated in 2005) (0/1)	−0.035	−0.041	0.001	−3.169	−4.176	0.004	0.001	0.003	0.343
	(0.050)	(0.125)	(0.014)	(3.298)	(5.623)	(0.007)	(0.048)	(0.009)	(0.579)
log(Requested amount)			0.036***	−0.013	−0.006	0.004	0.007	0.135**	0.773
			(0.011)	(0.010)	(0.014)	(0.018)	(0.026)	(0.057)	(0.818)
Personal collateral (0/1)	0.046	0.022	−0.027	0.033	0.021	−0.007	−0.009	0.030*	0.260
	(0.046)	(0.075)	(0.026)	(0.022)	(0.028)	(0.041)	(0.055)	(0.015)	(0.176)
Requested LTV			0.018***	0.037	0.001**	0.002	0.002	0.070***	0.028
			(0.006)	(0.037)	(0.000)	(0.008)	(0.010)	(0.019)	(0.021)
Requested LTV^2			0.038***	−0.043***	−0.004***	0.004	0.005	0.041*	0.008
			(0.005)	(0.021)	(0.001)	(0.038)	(0.055)	(0.021)	(0.007)
Experian business score	0.028***	0.047***	−0.064***		0.030***	−0.007***	−0.005	−0.141***	0.265
	(0.008)	(0.008)	(0.015)		(0.008)	(0.003)	(0.003)	(0.050)	(0.420)
Experian personal score	0.037	0.057***	−0.020	0.029*		−0.035***	−0.045***	−0.091*	0.182
	(0.058)	(0.018)	(0.034)	(0.017)		(0.004)	(0.004)	(0.048)	(0.136)
Loan officer fixed effects	No	No	No	No	No	No	No	No	No
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Month fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	14,916	14,916	14,916	14,916	14,916	14,916	14,916	14,916	14,916
Adj. R^2	0.05	0.07	0.19	0.11	0.10	0.73	0.20	0.07	0.06

Panel B: Originated loans in groups A and B in 2004

Dependent variable:	log(Originated amount) −log(Requested amount)	Personal collateral	Originated LTV −Requested LTV (%)	Interest rate	Experian business score	Experian personal score	Internal risk rating
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Group B (to be treated in 2005) (0/1)	0.020	0.325	−0.038	0.001	1.070	1.586	0.003
	(0.050)	(0.050)	(0.061)	(0.016)	(2.204)	(2.431)	(0.009)
log(Originated amount)				0.034***	−0.010	−0.006	0.003
				(0.010)	(0.009)	(0.012)	(0.015)
Personal collateral (0/1)	0.053	0.481	0.042	−0.023	0.032	0.018	−0.006
	(0.045)	(0.439)	(0.065)	(0.021)	(0.021)	(0.019)	(0.037)
Requested LTV		0.024		0.016***	0.033	0.001**	0.002
		0.021		(0.005)	(0.035)	(0.000)	(0.007)
Requested LTV^2		0.006		0.037***	−0.035***	−0.004***	0.004
		0.004		(0.005)	(0.022)	(0.001)	(0.032)
Experian business score	0.024***	0.198	0.034***	−0.055***		0.024***	−0.006***
	(0.008)	(0.242)	(0.008)	(0.012)		(0.007)	(0.002)
Experian personal score	0.041	0.167*	0.037***	−0.018	0.022		−0.032***
	(0.087)	(0.089)	(0.007)	(0.029)	(0.013)		(0.004)
Loan officer fixed effects	No	No	No	No	No	No	No
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Month fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	4740	4740	4740	4740	4740	4740	4740
Adj. R^2	0.07	0.05	0.11	0.21	0.12	0.10	0.69

The table compares the characteristics of applications and originated loans of Groups A and B in 2004. Panel A tests whether loan applications received by Group A (control) and Group B (to be treated in 2005) are different in the pretreatment period (2004). Panel B does the same for originated loans. All regressions are ordinary least squares regressions. Variables are defined in Appendix A. Standard errors are clustered at the loan officer level. Standard errors are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% level, respectively.

5.10 Table 8: Loan officer characteristics

Read: Table 8 shows heterogeneity by loan officer tenure and gender. More tenured loan officers and male loan officers respond more aggressively to the treatment. They are more likely to approve loans and approve larger loan amounts. These same characteristics are also associated with higher compensation in the treatment group.

Economic message: Incentives do not affect all agents equally. The response is stronger for officers who may have fewer career concerns or stronger competitive responses to performance-based pay.

5.11 Appendix B: Pre-treatment balance

Read: Appendix B compares Group A and Group B in 2004, before the treatment. The groups are statistically similar in application characteristics, originated-loan characteristics, and compensation.

Economic message: This is important for identification. If Group B was already more aggressive before treatment, the post-2005 results would be hard to interpret. The balance checks support the difference-in-differences design.

6 Short summary

- The paper studies a bank pilot that changed small-business loan officers from fixed salary to loan-prospecting incentives.
- Treatment was assigned through legacy HR systems, creating a credible difference-in-differences setting.

- Treated loan officers did not bring in more applications or observably better applications.
- Treated loan officers approved more loans and originated larger loans.
- Approved treated loans did not look worse using hard information such as credit scores, internal ratings, or interest rates.
- Treated loans defaulted substantially more often within 12 months.
- Approval decisions put more weight on hard information and less weight on soft information.
- The bank's old credit model lost predictive power in the treatment group, especially for aggressive loans.
- The pilot was discontinued after the risk-management division observed higher-than-expected defaults.

7 Conclusion

Agarwal and Ben-David show that loan prospecting incentives can weaken the informational advantage of relationship lending. The bank wanted loan officers to act more like salespeople and bring new business to the bank. Instead, treated loan officers increased lending by approving more applications, approving larger amounts, and moving faster through the process.

The most important result is that treated loans did not look worse on observable hard information but defaulted more often. This pattern points to a loss of soft information. Loan officers appear to have discounted unfavorable soft information that previously helped screen out risky borrowers.

The paper's deeper lesson is about organizational design. Incentives that reward measurable short-run output can undermine less measurable information production. In banking, this is especially dangerous because the quality of lending often depends on judgment, local information, and cautious use of soft signals.